

Question

CSE427 Class Test 4

Summer 2026

Duration: 30 Minutes

Name:

ID:

Sec:

SL:

Question [15 Points]

Let p and q be the second last and the last digit of your Student ID. For example, if someone's Student ID is 12345678, then $p = 7$ and $q = 8$. For your specific Student ID, these values will change.

An input image of size $(240 + p) \times (320 + q) \times 3$ is passed through the following CNN:

Phase	Details
1	The input image is convolved with 32 filters of size $7 \times 7 \times 3$, using stride = 2 and zero-padding = 3. The convolution output is then passed through a max-pooling layer with pool size = 3×3 , stride = 2, and zero-padding = 1.
2	The output of the first max-pooling layer is convolved with 64 filters of size 5×5 , using stride = 1 and zero-padding = 2. The convolution output is then passed through a max-pooling layer with pool size = 2×2 , stride = 2, and zero-padding = 0.
3	The output of the second max-pooling layer is convolved with 128 filters of size 3×3 , using stride = 2, and zero-padding = 1. No pooling operation is applied during this phase.
4	The output of the third phase is convolved with 64 filters of size 3×3 , using stride = 1 and zero-padding = 0. The convolution output is then passed through a max-pooling layer with pool size = 2×2 , stride = 2, and zero-padding = 0.

Now, determine the output height, width, and depth after each phase. Note that the max-pooling layers don't change the number of channels.

Tentative Solution

Assuming, $p = 6$ and $q = 5$.

Initial Input Image:

Input Height = $240 + 6 = 246$

Input Width = $320 + 5 = 325$

Input Depth = 3

Initial Image Size = $246 \times 325 \times 3$

Phase 1:

Step 1: Convolution

Filters = 32

Kernel = 7×7

Stride = 2

Padding = 3

Height:

$\text{floor}((246 + 2 \times 3 - 7) / 2) + 1 = \text{floor}((245) / 2) + 1 = 123$

Width:

$\text{floor}((325 + 2 \times 3 - 7) / 2) + 1 = \text{floor}((324) / 2) + 1 = 163$

After Convolution = $123 \times 163 \times 32$

Step 2: Max Pooling

Pool size = 3×3

Stride = 2

Padding = 1

Height:

$\text{floor}((123 + 2 \times 1 - 3) / 2) + 1 = \text{floor}((122) / 2) + 1 = 62$

Width:

$\text{floor}((163 + 2 \times 1 - 3) / 2) + 1 = \text{floor}((162) / 2) + 1 = 82$

Final output of Phase 1 = $62 \times 82 \times 32$

Phase 2:

Step 1: Convolution

Filters = 64

Kernel = 5×5

Stride = 1

Padding = 2

Height:

$$\text{floor}((62 + 2 \times 2 - 5) / 1) + 1 = \text{floor}((61) / 1) + 1 = 62$$

Width:

$$\text{floor}((82 + 2 \times 2 - 5) / 1) + 1 = \text{floor}((81) / 1) + 1 = 82$$

After Convolution = $62 \times 82 \times 64$

Step 2: Max Pooling

Pool size = 2×2

Stride = 2

Padding = 0

Height:

$$\text{floor}((62 + 2 \times 0 - 2) / 2) + 1 = \text{floor}((60) / 2) + 1 = 31$$

Width:

$$\text{floor}((82 + 2 \times 0 - 2) / 2) + 1 = \text{floor}((80) / 2) + 1 = 41$$

Final output of Phase 2 = $31 \times 41 \times 64$

Phase 3:

Convolution

Filters = 128

Kernel = 3×3

Stride = 2

Padding = 1

Height:

$$\text{floor}((31 + 2 \times 1 - 3) / 2) + 1 = \text{floor}((30) / 2) + 1 = 16$$

Width:

$$\text{floor}((41 + 2 \times 1 - 3) / 2) + 1 = \text{floor}((40) / 2) + 1 = 21$$

No pooling in this phase.

Final output of Phase 3 = $16 \times 21 \times 128$

Phase 4:

Step 1: Convolution

Filters = 64

Kernel = 3×3

Stride = 1

Padding = 0

Height:

$$\text{floor}((16 + 2 \times 0 - 3) / 1) + 1 = \text{floor}((13) / 1) + 1 = 14$$

Width:

$$\text{floor}((21 + 2 \times 0 - 3) / 1) + 1 = \text{floor}((18) / 1) + 1 = 19$$

After Convolution = $14 \times 19 \times 64$

Step 2: Max Pooling

Pool size = 2×2

Stride = 2

Padding = 0

Height:

$$\text{floor}((14 + 2 \times 0 - 2) / 2) + 1 = \text{floor}((12) / 2) + 1 = 7$$

Width:

$$\text{floor}((19 + 2 \times 0 - 2) / 2) + 1 = \text{floor}((17) / 2) + 1 = 9$$

Final output of Phase 4 = $7 \times 9 \times 64$